



Document Management System



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Title: Maternal & Congenital Toxoplasmosis: Diagnosis and Treatment

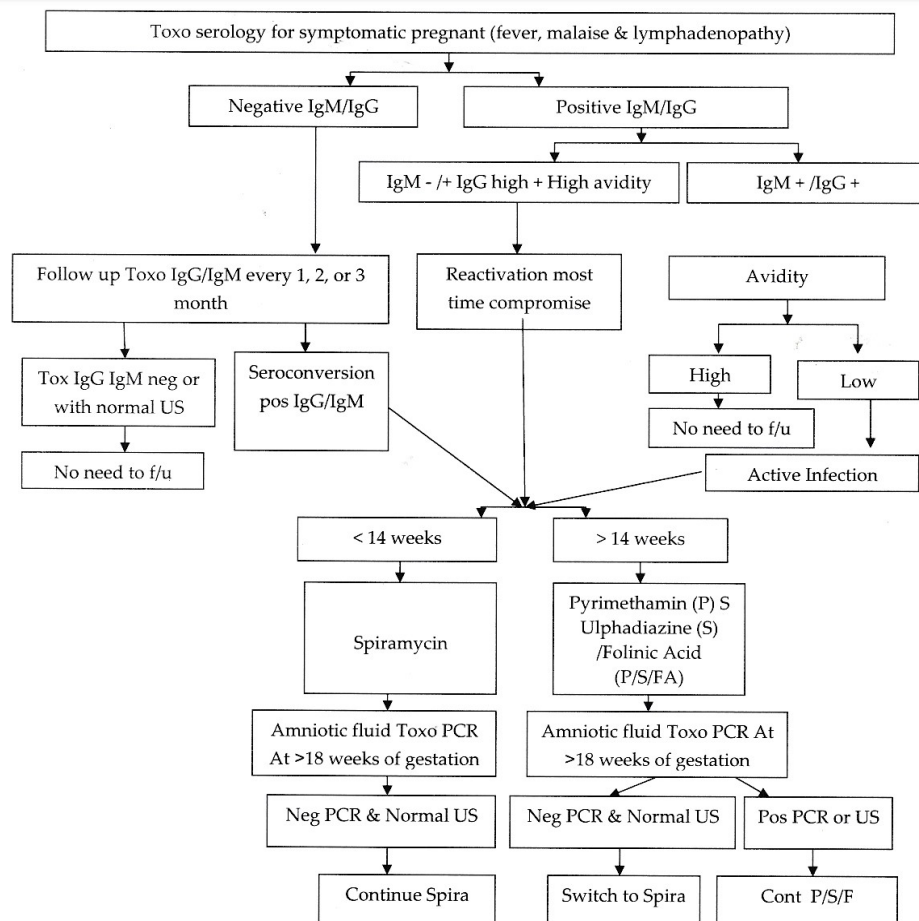
Introduction:

Toxoplasma gondii is an intracellular parasite can cause congenital disease as well. Congenital toxoplasmosis can occur when the infection acquired first time during pregnancy or, more rarely, shortly before conception. The risk of transmission essentially varies with gestational age and whether the mother receives treatment or not.The risk of mother child transmission increase from 7% in first trimesterto 24% in the second, to 59% in the third, while the severity decreases from 75% to a negligible risk in late pregnancy ^[1].The overall risk of transmission in mothers who have been treated during gestation is around 30%. ^[2]Fetuses who infected earlier in gestation generally have more severe disease.

Clinicalfeatures ^[1]:

- Premature birth/Low birth weight.
- Jaundice/Hepatosplenomegaly
- Chorioretinitis / retinal scarring.
- Intracranial calcification/ Hydrocephalus/Mental retardation.
- Pneumonia.
- Thrombocytopenia.
- Lymphadenopathy
- Myocarditis
- Ophthalmological assessment Retinal exam:chorioretinal scars or choroiditis andvirtuitis, uveitis, and cataracts.

Diagnosis: Criteria for evaluation of maternal serology.



1- Neonate:

A – Send Serum Toxo IgM & IgG (avidity testes not indicated here)

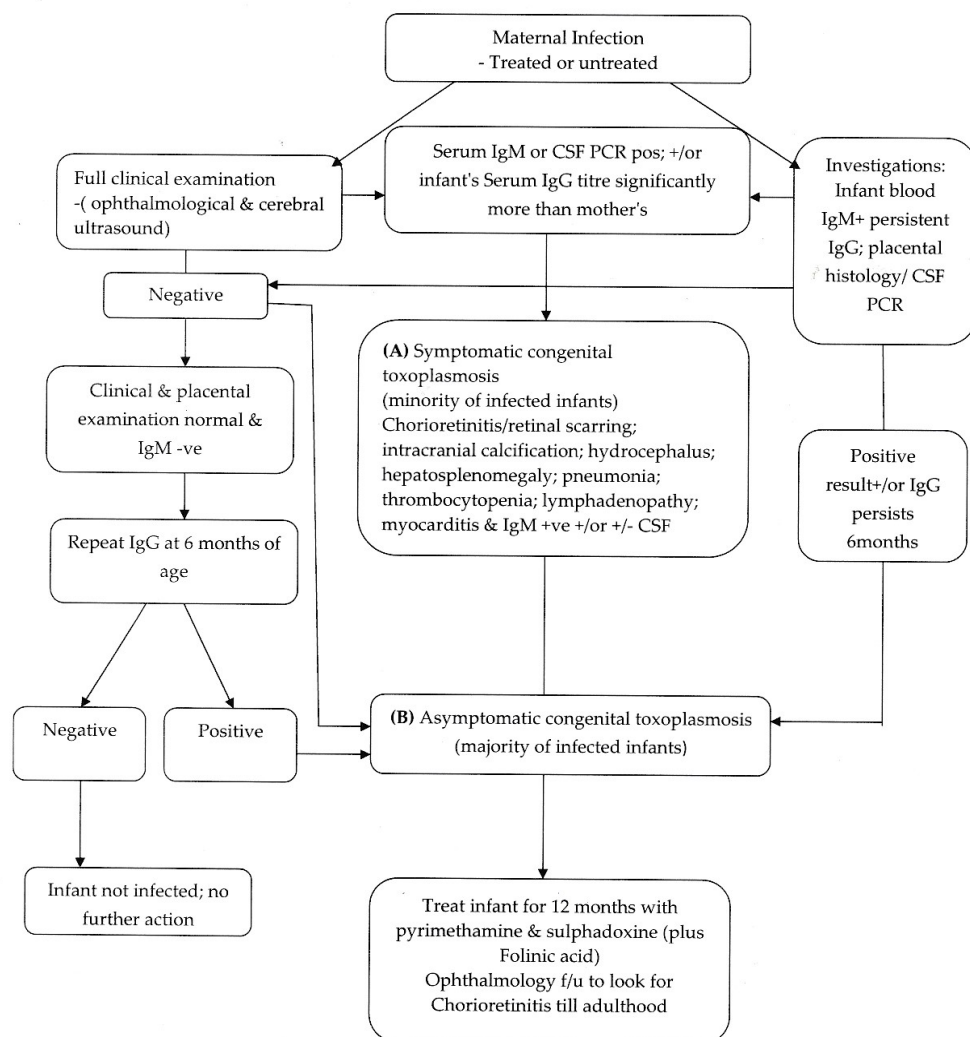
B- Send CSF Toxo PCR If neonate has CNS manifestation (At Royal Hospital Toxo PCR goes as referral test to France through biochemistry lab)

2- Mother:

Serum Toxo IgG (Not IgM or avidity testes indicated here)

*(Avidity test Percentagedepends on commercial kitused)

Management of infant at Risk of Toxoplasmosis



Prenatal treatment [4,5]:

1-Pregnant lady with suspected or documented infection (<18 wks of gestation) treatment with Sipramycin 1g (3 million U) every 8 hrs till delivery with low suspicion of fetal infection or those with documented negative result of amniotic fluid PCR and negative US finding.

2-Pregnant ladies > 18 wks of gestation (**Treatment to be continued till birth**)

A) Pyrimetahmine (loading dose 50mg/dose 12hrly for 2 days then 50mg /day as OD dose till delivery .

B) Sulfadiazine(loading dose: 75 mg/kg one dose) then 50mg /kg/dose 12 hrly (maximum 4 g/day) .

C) leucovorine 10 -20mg daily during and after 1 wk of Pyrimetahmine therapy

Treatment of neonate with congenital toxoplasmosis [4,5,6]:

-Pyrimethamine : Loading dose(2 mg/kg OD dose for 2 days), then 1 mg/kg OD for 6 month, then this dose every Monday, Wednesday, and Friday for other 6 months .

-Sulfadiazine (50 mg/kg/dose 12 hourly), and Leucovorin (5–10) mg per dose daily for 1 years and for 1 wk after pyrimethamine therapy.

- The use of corticosteroids is controversial and should be determined case by case but may be considered for active chorioretinitis or if CSF protein is > 1 gm/dL. Steroids are begun after loading doses of Pyrimethamine with Sulfadiazine have been administered. [3]
- **For later recurrences of active retinitis:** Azithromycin or Clindamycin are sometimes substituted for Sulfadiazine as second-line treatments, given with Pyrimethamine till resolution of active retinitis. Then these medicines may be useful without Pyrimethamine for suppression.

Prognosis:

Many children born with congenital toxoplasmosis who are treated in utero and throughout the first year of life have normal cognitive development and function well. Cognitive function was normal in 69% of treated infants compared with 14% of untreated infants or infants receiving short term therapy. No sensorineural hearing loss was observed in the treatment group compared with hearing deficits in 30% of controls

References

- 1-European Review for medical and Pharmacological Science congenital toxoplasmosis treatment D.Serranti,D.UONSESOP.valentini ,Department of Pediatric ,Catholic University of the Sacred Heart.
- 2-European Research Network on Congenital Toxoplasmosis, based on probability of infection.
- 3-Laboratory Diagnosis of Congenital Toxoplasmosis Christelle Pomares and Jose G. Montoya J Clin Microbiol. 2016 Oct; 54(10): 2448–2454. May 4. doi: 10.1128/JCM.00487-16.
- 4-Management of Congenital Toxoplasmosis Rima McLeod • Joseph Lykins • A. Gwendolyn Noble • Peter Rabiah • Charles N. Swisher • Peter T. Heydemann • David McLone • David Frim • Shawn Withers • Fatima Clouser • Kenneth Boyer. Curr Pediatr Rep (2014) 2:166–194 DOI 10.1007/s40124-014-0055-7.
- 5-Clinical Guideline Toxoplasmosis in pregnancy Policy developed by: SA Maternal & Neonatal Clinical Network Approved SA Health Safety & Quality Strategic Governance Committee on: 24 June 2015 Next review due: 24 June 2018.
- 6-Guideline for treatment of T.gondi infection in the pregnant women. (Daffos F, Capella _ Pavlovsky Metal . Prenatal management of 746 pregnancies at risk for congenital toxoplasmosis . N Eng J Med 1988; 31: 271-275).

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Review Details

	Review BY	Review Date
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